Markov Chains and Metropolis–Hastings

From Markov's letters to sampling arbitrary distributions

Anton Selitskiy

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Why Markov chains?

- Bayesian inference needs expectations under a posterior $\pi(\theta) \propto p(\theta)\, p(\mathcal{D} \mid \theta)$, known only up to a constant.
- Independent samples from π are usually out of reach (importance sampling degenerates in high dimension).
- **Idea of MCMC:** build a random sequence $X_0, X_1, X_2, \ldots$ that is *not* independent, but whose long-run behavior is governed by π:

$$\frac{1}{n} \sum_{k=1}^{n} f(X_k) \;\longrightarrow\; \mathbb{E}_\pi f.$$

- Today:
  1. Markov chains on a finite state space: stationary distribution, detailed balance, ergodic theorem, burn-in, independence.
  2. The same ideas on a continuous state space.
  3. Metropolis–Hastings: a chain with *any* prescribed stationary distribution.

Why Markov chains?

	discrete state (E countable)	**continuous state** ($E = \mathbb{R}^d$)
discrete time $n \in \mathbb{N}$	**Markov chain.** Transition matrix P_{xy}. $\mu_{n+1}(y) = \sum_x \mu_n(x)\, P_{xy}, \quad P^{(n)} = P^n$ Stationary: $\pi P = \pi$. *Examples:* simple random walk on $\mathbb{Z}$; Ehrenfest urn; Galton–Watson branching; card shuffling; Metropolis chain on a finite graph.	**Chain on a general state space.** Kernel $K(x, \mathrm{d}y)$, with density $k(x, y)$ when one exists. $p_{n+1}(y) = \int p_n(x)\, k(x, y)\, \mathrm{d}x$ Stationary: $\pi K = \pi$. *Examples:* Gaussian random walk $X_{n+1} = X_n + \sigma\xi_n$; AR(1) $X_{n+1} = aX_n + \sigma\xi_n$; Kalman state-space models; MCMC; DDPM chains.
continuous time $t \geq 0$	**Jump process.** Rate matrix Λ, $P(t) = e^{t\Lambda}$. $\frac{\mathrm{d}p_y}{\mathrm{d}t} = \sum_x p_x \Lambda_{xy}$ (master / forward equation) Stationary: $\pi\Lambda = 0$.	**Diffusion.** SDE $\mathrm{d}X_t = b(X_t)\,\mathrm{d}t + g(X_t)\,\mathrm{d}W_t$. $\partial_t p = -\nabla \cdot (bp) + \frac{1}{2}\sum_{i,j} \partial_i \partial_j \big((gg^\top)_{ij}\, p\big)$ (Fokker–Planck) Stationary: ${}^*\pi = 0$.

Markov's chain

Is independence necessary for the law of large numbers?

- Classical LLN: if $X_1, X_2, \ldots$ are i.i.d., then $\frac{1}{n}\sum_k X_k \to \mathbb{E}X_1$. The joint law factorizes,

$$p(x_1, \ldots, x_n) = p(x_1)\, p(x_2) \cdots p(x_n).$$

- Early 1900s: P. A. Nekrasov (Moscow) claimed that independence is *necessary*.
- A. A. Markov disagreed. 1906: an LLN for variables that depend on their predecessor.
- 1913: an illustration on real data, the first 20 000 letters of Pushkin's *Eugene Onegin*.

Markov's experiment (1913)

Label every letter as a vowel (V) or a consonant (C).

	Markov 20 000 letters	first stanza 282 letters
$P(V)$	0.432	0.418
$P(V \mid \text{previous } V)$	0.128	0.043
$P(V \mid \text{previous } C)$	0.663	0.689

If letters were independent, all three rows would be equal.

Moj djadja samykh chestnykh pravil...

CVCCVCVCVCVCCVCCCVCCCVCV
CCVCCVCVCCVCCVCVCVCVCVCV...

χ^2 test on the 2×2 table of pairs of the stanza: $p \approx 10^{-27}$.

Letters are strongly *dependent*, yet their frequencies stabilize.

The simplest model of dependence

Replace the product of marginals by a product of *transition functions*:

$$p(x_0, x_1, \dots, x_n) = p_0(x_0)\, p_1(x_0, x_1)\, p_2(x_1, x_2) \cdots p_n(x_{n-1}, x_n),$$

$$\sum_x p_0(x) = 1, \qquad \sum_y p_k(x, y) = 1 \ \text{ for every } x.$$

- A valid distribution: sum out $x_n, x_{n-1}, \dots$ one at a time; each sum equals 1.
- i.i.d. is the special case $p_k(x, y) = q(y)$.
- Coordinates $\xi_k(x_0, \dots, x_n) = x_k$ ("the k-th letter") satisfy

$$P(\xi_{k+1} = a_{k+1} \mid \xi_k = a_k, \dots, \xi_0 = a_0) = P(\xi_{k+1} = a_{k+1} \mid \xi_k = a_k) = p_{k+1}(a_k, a_{k+1}).$$

Markov property

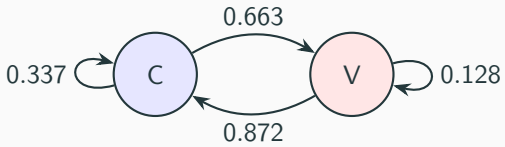
Given the present, the future does not depend on the past:
$P\big(\xi_{k+1} = a \mid \sigma(\xi_0, \dots, \xi_k)\big) = P(\xi_{k+1} = a \mid \xi_k)$.

Transition matrix

Time-homogeneous chain on states $\{1, 2, \dots\}$: $p_{ij} = P(X_{n+1} = j \mid X_n = i)$ form a **stochastic matrix** P ($p_{ij} \geq 0$, rows sum to 1).

$$P = \begin{pmatrix} p_{CC} & p_{CV} \\ p_{VC} & p_{VV} \end{pmatrix} = \begin{pmatrix} 0.337 & 0.663 \\ 0.872 & 0.128 \end{pmatrix}$$

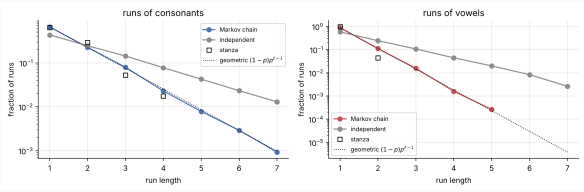
- Runs of a letter type are geometric: a vowel run continues with probability 0.128.
- The stanza follows the chain, not the independent model.



Simulated text: chain vs independent letters

```
Markov chain :  CVCVCCVCCCVCCCCVCVCVCVCVCVCVCVCCVCCVCVCVCCVCVC
independent  :  VVVVCCCCCVCCVCCCCCVCCCVCVCVCVCVCVCCCVVCVVVVCVCCV
real stanza  :  CVCCVCVCVCVCCVCCCVCCCVCVCCVCCVCVCCVCCVCVCVCVCVCV
```

Three sequences with the *same* vowel rate 0.43: Markov's chain, independent letters, the actual stanza.



A **run** is a maximal block of equal letters: `C VV CCC` has runs of length 1, 2, 3.

- x-axis: run length ℓ; y-axis (log): fraction of runs of that length.
- Colored: Markov chain. Gray: independent letters. Squares: the stanza.
- Dotted: $P(\text{run} = \ell) = (1-p)p^{\ell-1}$ with $p =$ probability to repeat the letter ($p_{CC} = 0.337$, $p_{VV} = 0.128$).
- Independent letters produce long runs (`VVVV`); the chain and the stanza almost never repeat a vowel.

Evolution of distributions

Chapman–Kolmogorov: two steps = sum over the letter in between

$P_{ij}^{(2)} = P(X_2 = j \mid X_0 = i)$. To go from i to j in two steps, the middle letter is either C or V:

$$P_{VV}^{(2)} = \underbrace{p_{VC}\, p_{CV}}_{V \to C \to V} + \underbrace{p_{VV}\, p_{VV}}_{V \to V \to V} = 0.872 \cdot 0.663 + 0.128 \cdot 0.128 = 0.594.$$

This is the (V, V) entry of $P \cdot P$. In general, splitting $m + n$ steps at time m:

$$\boxed{P_{ij}^{(m+n)} = \sum_k P_{ik}^{(m)}\, P_{kj}^{(n)}} \qquad \Longrightarrow \qquad P^{(n)} = P^n.$$

- Each term: probability of being at k after m steps, times probability of going from k to j in the next n steps. The Markov property lets us multiply: after reaching k, the past is forgotten.
- Meaning of 0.594: two letters after a vowel, a vowel is *more* likely than average (0.432), because of alternation.

Forward and backward equations

Forward: how the law of the current letter evolves. Let $\mu_n(j) = P(X_n = j)$. Letter $n+1$ is a vowel if (letter n is C *and* we switch) or (letter n is V *and* we repeat):

$$\mu_{n+1}(V) = \mu_n(C)\, p_{CV} + \mu_n(V)\, p_{VV}, \qquad \text{in matrix form} \quad \mu_{n+1} = \mu_n P \quad (\mu \text{ a row vector}).$$

Backward: how a prediction depends on the start. For a function f of the state, let $u_n(i) = \mathbb{E}[f(X_n) \mid X_0 = i]$. Condition on the *first* step instead of the last:

$$u_{n+1}(i) = \sum_j p_{ij}\, u_n(j), \qquad \text{in matrix form} \quad u_{n+1} = P u_n, \quad u_0 = f \quad (u \text{ a column vector}).$$

Example: $f = \mathbf{1}\{V\}$ gives $u_n(i) = P(\text{letter } n \text{ is } V \mid \text{letter 0 is } i)$.

Both are P^n read from different sides:
$\mathbb{E}[f(X_n)] = \sum_j \mu_n(j) f(j) = \mu_0 P^n f = \sum_i \mu_0(i) u_n(i)$.
P acts on distributions from the left (forward) and on functions from the right (backward).

Two states: everything is explicit

Derivation. Use $\mu_n(C) = 1 - \mu_n(V)$ in the forward equation:

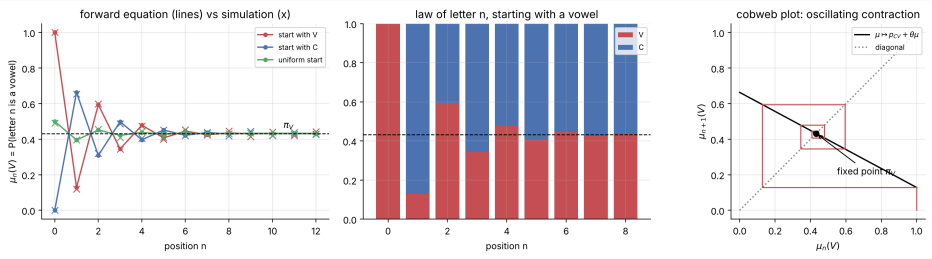
$$\mu_{n+1}(V) = \big(1 - \mu_n(V)\big)p_{CV} + \mu_n(V)\,p_{VV} = p_{CV} + \underbrace{(p_{VV} - p_{CV})}_{=\,\theta}\,\mu_n(V).$$

- $\theta = p_{VV} - p_{CV} = 1 - p_{VC} - p_{CV} =$ (chance of a vowel after a vowel) $-$ (after a consonant): the **memory** of the chain. $\theta = 0$ means independent letters. It is also the second eigenvalue of P.
- The fixed point solves $\pi_V = p_{CV} + \theta\pi_V$. Subtracting: $\mu_{n+1}(V) - \pi_V = \theta\,\big(\mu_n(V) - \pi_V\big)$, hence

$$\boxed{\mu_n(V) = \pi_V + \theta^n\big(\mu_0(V) - \pi_V\big)}.$$

  The distance to π_V is multiplied by θ at every letter.
- Markov's chain: $\theta = 0.128 - 0.663 = -0.535$. Since $\theta < 0$ the sign of the deviation flips at each step: the law *oscillates* while converging (a vowel makes the next letter likely a consonant, which makes the one after likely a vowel, ...).

Two states: oscillating convergence



- **Left:** $\mu_n(V)$ for three first letters (V, C, 50/50). Lines: the exact formula; crosses: fraction of vowels at position n in 5000 simulated texts.
- **Middle:** the law of letter n as stacked bars (red $= V$, blue $= C$), starting with a vowel; dashed line π_V.
- **Right:** cobweb plot of the map $\mu \mapsto p_{CV} + \theta\mu$: go up to the line, across to the diagonal, repeat. The slope $\theta < 0$ makes a square spiral into the fixed point π_V.

Stationary distribution

Definition

π is **stationary** if $\pi P = \pi$: if $X_0 \sim \pi$, then $X_n \sim \pi$ for all n.

Two states. The V-component of $\pi P = \pi$ reads

$$\pi_V = \pi_C\, p_{CV} + \pi_V\, p_{VV} \iff \pi_V(1 - p_{VV}) = \pi_C\, p_{CV} \iff \underbrace{\pi_V\, p_{VC}}_{\text{flow } V \to C} = \underbrace{\pi_C\, p_{CV}}_{\text{flow } C \to V}.$$

In equilibrium, as much probability leaves V per step as enters it. With $\pi_C + \pi_V = 1$:

$$\pi = \left(\frac{p_{VC}}{p_{CV} + p_{VC}},\ \frac{p_{CV}}{p_{CV} + p_{VC}}\right) = \left(\frac{0.872}{1.535},\ \frac{0.663}{1.535}\right) = (0.568,\ 0.432).$$

- $\pi_V = 0.432$: **exactly Markov's fraction of vowels.** A model built from *pairs* reproduces the frequency of *single* letters. Why? (next slides)
- It is also the fixed point of the previous slide: $\mu_n(V) \to \pi_V$ from any start.

Why π equals the letter frequencies (1): counting

Count in the stanza: 282 letters, $N_V = 118$, $N_C = 164$, and the 281 pairs

	next C	next V
after C	$N_{CC} = 51$	$N_{CV} = 113$
after V	$N_{VC} = 112$	$N_{VV} = 5$

- Every vowel except possibly the first letter is the *second* letter of exactly one pair: $N_V = N_{CV} + N_{VV}$, $118 = 113 + 5$.
- The estimated transition probabilities are $\hat{p}_{CV} = N_{CV}/N_C$ and $\hat{p}_{VV} = N_{VV}/N_V$, so

$$N_V = N_C\,\hat{p}_{CV} + N_V\,\hat{p}_{VV} \quad \xRightarrow{\div n} \quad \hat{f}_V = \hat{f}_C\,\hat{p}_{CV} + \hat{f}_V\,\hat{p}_{VV},$$

  which is exactly the stationarity equation, up to an error $O(1/n)$ from the first and last letters.
- **Conclusion:** the letter frequencies $\hat{f}$ *are* the stationary distribution of the estimated $\hat{P}$, for any text. So Markov's 0.432 from pairs *had* to match his 0.432 from single letters.

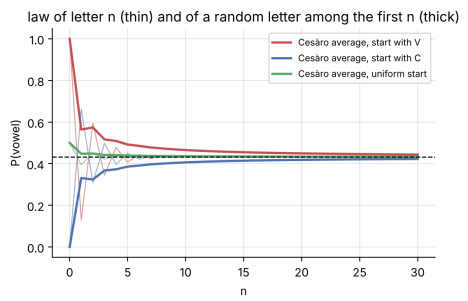
Why π equals the letter frequencies (2): averaging

Cesàro average. The average of the first n terms of a sequence, $\frac{a_0+a_1+\cdots+a_{n-1}}{n}$. If $a_k \to a$, then the averages also tend to a (early terms are diluted).

- Take $a_k = \mu_k(V) = P(\text{letter } k \text{ is } V)$.
- The *expected* fraction of vowels among the first n letters is

$$\mathbb{E}\Big[\frac{1}{n}\sum_{k<n}\mathbf{1}\{X_k = V\}\Big] = \frac{1}{n}\sum_{k<n}\mu_k(V),$$

  a Cesàro average. It is also $P(X_K = V)$ for a uniformly random position K.
- Since $\mu_k(V) \to \pi_V$, the expected fraction $\to \pi_V$, whatever the first letter.



Thin: $\mu_n(V)$. Thick: Cesàro averages. Three first letters, same limit π_V.

This is about the *expected* frequency. That the frequency in *one* text converges is Markov's theorem (next slide).

Markov's law of large numbers (1906)

Theorem (two states, $0 < p_{ij} < 1$)

Let $S_n = \sum_{k=1}^{n} \mathbf{1}\{X_k = V\}$ be the number of vowels among n letters. Then for every $\varepsilon > 0$,

$$P\Big(\Big|\frac{S_n}{n} - \pi_V\Big| > \varepsilon\Big) \;\longrightarrow\; 0,$$

although the letters are *not* independent.

Proof (Chebyshev's method, as Markov did). Start in equilibrium for simplicity. Then $\mathrm{Cov}(\mathbf{1}\{X_i = V\}, \mathbf{1}\{X_j = V\}) = \pi_V \pi_C \, \theta^{|i-j|}$: correlations decay geometrically. Hence

$$\mathrm{Var}\left(\frac{S_n}{n}\right) = \frac{\pi_V \pi_C}{n^2} \sum_{i,j=1}^{n} \theta^{|i-j|} \;\le\; \frac{\pi_V \pi_C}{n} \cdot \frac{1 + |\theta|}{1 - |\theta|} \;\longrightarrow\; 0,$$

and Chebyshev's inequality $P(|S_n/n - \pi_V| > \varepsilon) \le \mathrm{Var}(S_n/n)/\varepsilon^2$ finishes the proof. A different start changes $\mathbb{E}[S_n/n]$ only by $O(1/n)$ (previous slide).

The point: independence is not needed, only that dependence *fades* ($|\theta| < 1$). This is the seed of the ergodic theorem behind MCMC.

Detailed balance and ergodicity

Detailed balance

In stationarity, the flow from i to j per step is $F_{ij} = \pi_i p_{ij}$.

- **Global balance** ($\pi P = \pi$): for each state, total inflow = total outflow.
- **Detailed balance:** $\pi_i p_{ij} = \pi_j p_{ji}$ for *every pair*: the flow matrix F is symmetric.

Proposition: detailed balance $\Rightarrow$ π is stationary

Sum over i and use $\sum_i p_{ji} = 1$: $\quad (\pi P)_j = \sum_i \pi_i p_{ij} = \sum_i \pi_j p_{ji} = \pi_j.$

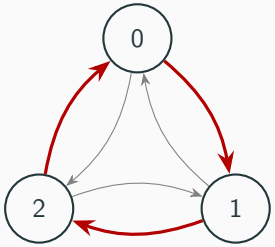
- **Sufficient and easy to check:** one pair of states at a time, no sum over all states. This is the condition Metropolis–Hastings is built to satisfy. Not necessary: see the next slide.
- *Meaning:* each transition is balanced by its reverse; the stationary chain looks the same played backwards (*reversibility*). Origin: statistical physics (Boltzmann, Einstein, Onsager); in MCMC since Metropolis et al. (1953).
- Two states: detailed balance $\equiv$ stationarity, so every two-state chain is reversible.

Stationary but not reversible

Cycle on $\{0, 1, 2\}$: forward with $p = 0.8$, backward with 0.2:

$$P = \begin{pmatrix} 0 & 0.8 & 0.2 \\ 0.2 & 0 & 0.8 \\ 0.8 & 0.2 & 0 \end{pmatrix}.$$

- Uniform π is stationary (columns also sum to 1).
- But $\pi_0 p_{01} = 0.267 \neq 0.067 = \pi_1 p_{10}$: net circulation $0 \to 1 \to 2 \to 0$.
- Played backwards: forward steps $80\% \to 20\%$.



Kolmogorov's criterion: reversible $\iff$ around every cycle $p_{01}p_{12}p_{20} = p_{02}p_{21}p_{10}$.
Here $0.512 \neq 0.008$.

The ergodic theorem (finite state space)

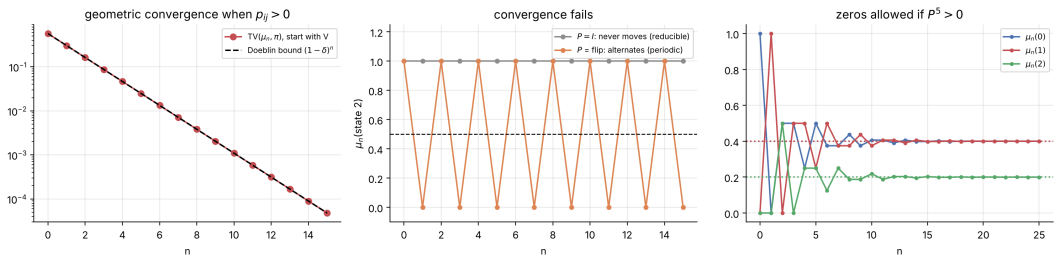
Theorem

Let P be **irreducible** (every state reaches every state) and **aperiodic** (no cyclic structure); equivalently, $P^m > 0$ entrywise for some m. Then

1. there is a unique stationary distribution π;
2. $\mu_0 P^n \to \pi$ geometrically fast, for every μ_0;
3. for every f, $\frac{1}{n}\sum_{k=1}^{n} f(X_k) \to \sum_x f(x)\pi(x)$ almost surely (LLN).

- Simplest sufficient condition: $p_{ij} > 0$ for all i, j.
- **Doeblin:** with $\delta = \sum_j \min_i p_{ij} > 0$, $\quad \|\mu P - \nu P\|_{\mathrm{TV}} \le (1-\delta)\|\mu - \nu\|_{\mathrm{TV}}$.
- Markov's chain: $\delta = 0.337 + 0.128 = 0.465$, contraction $0.535 = |\theta|$ (sharp).

Convergence and how it fails

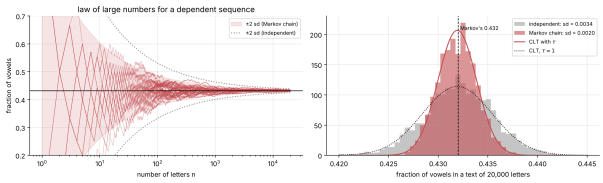


- $P = I$: never moves (*reducible*); every distribution is stationary.
- $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$: alternates (*periodic*); μ_n oscillates forever, but time averages still converge.
- Zero entries are fine as long as $P^m > 0$ for some m.

Markov's LLN: dependence changes only the error bar

$$\mathrm{sd}\left(\frac{1}{n}\sum_k \mathbf{1}\{X_k = V\}\right) \approx \sqrt{\frac{\pi_V \pi_C\, \tau}{n}}, \qquad \tau = 1 + 2\sum_{k\geq 1}\rho_k = \frac{1+\theta}{1-\theta} = 0.30$$

(ρ_k = correlation between letters k apart; τ = **integrated autocorrelation time**).



Effective sample size

For independent samples, Var(average) $= \sigma^2/n$. For a chain it is $\sigma^2\tau/n = \sigma^2/\mathrm{ESS}$ with $\boxed{\mathrm{ESS} = n/\tau}$: n correlated samples are as informative about a mean as ESS independent ones. Markov's 20 000 letters: $\mathrm{ESS} \approx 66\,000$ ($\theta < 0$ helps). In MCMC usually $\tau > 1$ and $\mathrm{ESS} < n$.

Burn-in

A chain started at a fixed state is not yet distributed as π. Discard the first n_{burn} states, with

$$\|\mu_n - \pi\|_{\text{TV}} = |\theta|^n \|\mu_0 - \pi\|_{\text{TV}} < \varepsilon \iff n > \frac{\log(\|\mu_0 - \pi\|_{\text{TV}}/\varepsilon)}{\log(1/|\theta|)}.$$

Markov's chain, start with V: $n_{\text{burn}} = 3,\ 7,\ 11$ for $\varepsilon = 10^{-1}, 10^{-2}, 10^{-3}$.

Bias of the vowel frequency, texts starting with V:

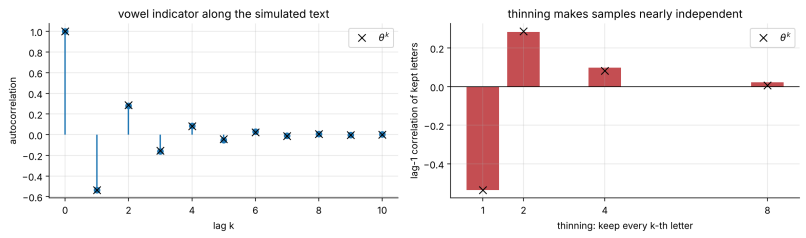
length	no burn-in	burn-in 8	sd
10	0.036	0.000	0.085
100	0.004	0.000	0.027
1000	0.0004	0.000	0.009

- Bias of an average from the start is $O(1/n)$; the noise is $O(1/\sqrt{n})$.
- Burn-in matters for **short runs** and **bad starts** (far tails, other modes).
- In practice θ is unknown: trace plots, several chains from dispersed starts ($\hat{R}$).

Independent samples

Consecutive states are correlated: here $\mathrm{corr}(\mathbf{1}\{X_k = V\}, \mathbf{1}\{X_{k+j} = V\}) = \theta^j$.

1. **Keep everything**, report $\mathrm{ESS} = n/\tau$ (standard for averages).
2. **Thinning**: keep every k-th state; correlation θ^k. Saves memory, never improves averages.
3. **Independent chains**: last state of each of M chains; exactly independent, one burn-in per sample.
4. **Exact methods**: regeneration, coupling from the past (special cases).



Continuous state space

From a matrix to a kernel

matrix entry $p_{ij} \longrightarrow$ *transition density* $k(x, y)$, *sum over* $j \longrightarrow$ *integral over* y

Markov kernel

$K(x, A) = P(X_{n+1} \in A \mid X_n = x)$: for each x a probability measure, measurable in x. Often $K(x, \mathrm{d}y) = k(x, y)\,\mathrm{d}y$, with $\int k(x, y)\,\mathrm{d}y = 1$.

Running example: Gaussian AR(1) chain

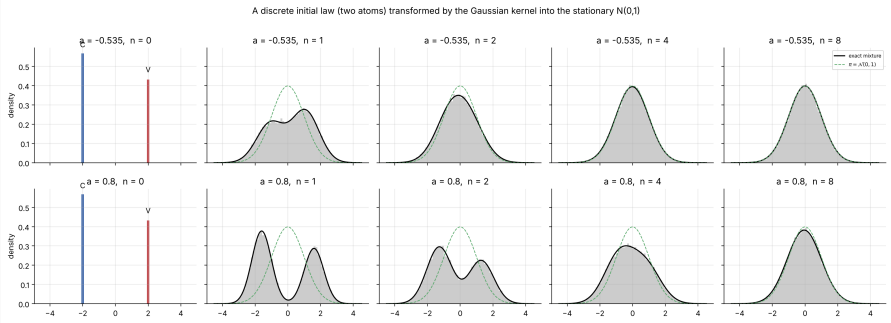
$$X_{n+1} = a\,X_n + \sqrt{1 - a^2}\,\xi_{n+1}, \qquad \xi_n \sim \mathcal{N}(0, 1), \qquad k(x, y) = \mathcal{N}\big(y;\, ax,\, 1 - a^2\big).$$

- a plays the role of θ; $k(x, y) > 0$ everywhere plays the role of $p_{ij} > 0$.
- Chapman–Kolmogorov: $k^{(n)}(x, y) = \int \cdots \int = \mathcal{N}\big(y;\, a^n x,\, 1 - a^{2n}\big)$.

Forward equation: transformation of the initial law

$$p_{n+1}(y) = \int p_n(x)\, k(x,y)\, \mathrm{d}x. \qquad \text{Start: } X_0 = \pm 2 \text{ with prob. } 0.432,\ 0.568 \text{ (“V” and “C”).}$$

$$\mu_n = 0.432\,\mathcal{N}(2a^n, 1-a^{2n}) + 0.568\,\mathcal{N}(-2a^n, 1-a^{2n}) \longrightarrow \mathcal{N}(0,1).$$



Stationarity, detailed balance, ergodicity on $\mathbb{R}$

- **Stationary:** $\int \varphi(x)k(x,y)\,\mathrm{d}x = \varphi(y)$ for $\varphi = \mathcal{N}(0,1)$ (fixed point of $v \mapsto a^2 v + 1 - a^2$).
- **Detailed balance:** $\varphi(x)k(x,y) = \varphi(y)k(y,x)$: here a bivariate Gaussian with correlation a, symmetric in (x,y).
- **Detailed balance $\Rightarrow$ stationary** (same proof as before, integral instead of sum): $\int \pi(x)k(x,y)\,\mathrm{d}x = \int \pi(y)k(y,x)\,\mathrm{d}x = \pi(y)$. For kernels with atoms: $\pi(\mathrm{d}x)K(x,\mathrm{d}y) = \pi(\mathrm{d}y)K(y,\mathrm{d}x)$ as measures.
- **Non-reversible example:** $X_{n+1} = aR_\phi X_n + \sqrt{1-a^2}\,\xi$ on $\mathbb{R}^2$ (rotation): $\mathcal{N}(0, I)$ stationary, probability circulates.

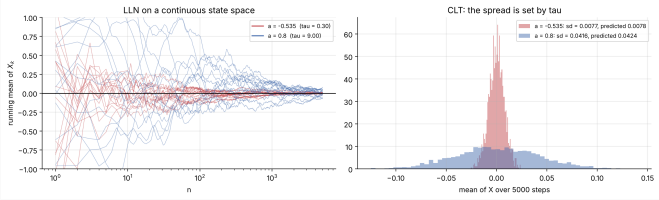
Ergodic theorem (general state space; Tierney 1994)

If (i) $\pi K = \pi$, (ii) K is π-**irreducible** (every set A with $\pi(A) > 0$ is reached with positive probability from every x), and (iii) K is **aperiodic**, then $\|K^n(x,\cdot) - \pi\|_{\mathrm{TV}} \to 0$ for π-a.e. x (every x if Harris recurrent), and $\frac{1}{n}\sum f(X_k) \to \mathbb{E}_\pi f$ a.s.

Sufficient for (ii)+(iii): π has a positive density and $K(x,\mathrm{d}y) \geq c(x,y)\,\mathrm{d}y$ with $c > 0$ (every set reached in *one* step).

LLN and CLT on $\mathbb{R}$

τ depends on the statistic: $f(x) = x$: $\tau = \frac{1+a}{1-a}$; $f(x) = x^2$: $\tau = \frac{1+a^2}{1-a^2}$.



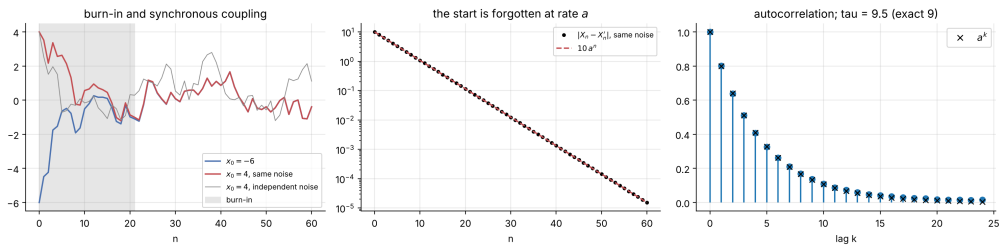
Effective sample size $\mathrm{ESS} = n/\tau$ (number of independent samples with the same accuracy for the mean):

- $a = -0.535$ (Markov's θ): $\tau = 0.30$, so 5000 steps are worth $\approx 16\,500$ independent samples;
- $a = 0.8$: $\tau = 9$, so 1000 steps are worth only ≈ 111; the histogram of means is $3\times$ wider than for independent samples.

Burn-in and independence on $\mathbb{R}$: coupling

No exact "refresh" on a continuous space. Instead, **synchronous coupling**: two chains, different starts, *same* noise:

$$X_n - X'_n = a^n(x_0 - x'_0) \quad \Longrightarrow \quad W_2(\mu_n, \pi) \le a^n \, W_2(\mu_0, \pi).$$



Burn-in from $x_0 = -6$: bias $a^n x_0 < 0.05$ needs $n \approx 21$. Thinning by k: correlation a^k.

Dictionary

	Markov's two-state chain	Gaussian AR(1) chain
transitions	p_{ij}, $\sum_j p_{ij} = 1$	$k(x,y) = \mathcal{N}(y; ax, 1-a^2)$, $\int k \, \mathrm{d}y = 1$
n steps	$P^n = \Pi + \theta^n R$	$\mathcal{N}(y; a^n x, 1-a^{2n})$
forward	$\mu_{n+1} = \mu_n P$	$p_{n+1}(y) = \int p_n(x) k(x,y) \, \mathrm{d}x$
backward	$u_{n+1} = P u_n$	$u_{n+1}(x) = \int k(x,y) u_n(y) \, \mathrm{d}y$
stationary	$\pi = (0.568, 0.432)$	$\mathcal{N}(0,1)$
detailed balance	$\pi_i p_{ij} = \pi_j p_{ji}$	$\varphi(x)k(x,y) = \varphi(y)k(y,x)$
convergence	$\lvert\theta\rvert^n$, Doeblin	a^n, coupling
autocorrelation	θ^k, $\tau = \frac{1+\theta}{1-\theta}$	a^k, $\tau_x = \frac{1+a}{1-a}$

So far: *chain given* $\rightarrow$ compute π. MCMC: π *given* $\rightarrow$ construct the chain.

Metropolis–Hastings

The inverse problem

- Given a target $\pi(x) \propto w(x)$, with w computable and the constant $Z = \int w$ unknown.
- Wanted: a kernel K with $\pi K = \pi$, irreducible and aperiodic. Then the ergodic theorem gives $\frac{1}{n}\sum f(X_k) \to \mathbb{E}_\pi f$.
- **Detailed balance is the easy route:** a *local* condition (one pair at a time) that needs only ratios $\pi(y)/\pi(x) = w(y)/w(x)$.

Metropolis–Hastings (Metropolis et al. 1953; Hastings 1970)

From the current state x:

1. propose $y \sim q(y \mid x)$;
2. accept with probability $\alpha(x, y) = \min\left\{1,\ \dfrac{w(y)\, q(x \mid y)}{w(x)\, q(y \mid x)}\right\}$; otherwise stay at x.

Symmetric proposal ($q(y \mid x) = q(x \mid y)$): $\alpha = \min\{1, w(y)/w(x)\}$ (Metropolis).

Why it works: the plan

We want the ergodic theorem for the MH chain, so we check its three hypotheses:

hypothesis	how we check it	where
(i) $\pi K = \pi$	**detailed balance** $\Rightarrow$ stationarity	next slide
(ii) irreducible	proposal reaches every set, target > 0	slide after
(iii) aperiodic	positive probability to reach a set in one step (or to stay)	slide after

Then: $\frac{1}{n}\sum_{k=1}^{n} f(X_k) \to \mathbb{E}_\pi f$ for every f with $\mathbb{E}_\pi|f| < \infty$, from any start.

Detailed balance is the key design idea: it is a condition on *pairs* (x, y) and involves π only through the ratio $\pi(y)/\pi(x) = w(y)/w(x)$, so the unknown constant Z never appears.

Step (i): MH satisfies detailed balance

Discrete states. For $y \neq x$, $p_{xy} = q(y \mid x)\,\alpha(x, y)$. Then

$$\pi(x)\,p_{xy} = \frac{w(x)}{Z}\,q(y \mid x)\min\left\{1, \frac{w(y)q(x \mid y)}{w(x)q(y \mid x)}\right\} = \frac{1}{Z}\min\left\{w(x)q(y \mid x),\ w(y)q(x \mid y)\right\},$$

which is **symmetric in x and y**, so $\pi(x)p_{xy} = \pi(y)p_{yx}$. For $y = x$ (rejection, p_{xx}) there is nothing to check. By the Proposition (detailed balance $\Rightarrow$ stationarity), $\pi P = \pi$.

Continuous states. The MH kernel is

$$K(x, \mathrm{d}y) = \underbrace{q(y \mid x)\,\alpha(x, y)\,\mathrm{d}y}_{\text{accepted move}} + \underbrace{r(x)\,\delta_x(\mathrm{d}y)}_{\text{rejection: stay}}, \qquad r(x) = 1 - \int q(y \mid x)\alpha(x, y)\,\mathrm{d}y.$$

- Density part: $\pi(x)q(y \mid x)\alpha(x, y) = \frac{1}{Z}\min\{w(x)q(y \mid x), w(y)q(x \mid y)\}$, symmetric, as above.
- Atom part: $\pi(\mathrm{d}x)\,r(x)\,\delta_x(\mathrm{d}y)$ lives on the diagonal $y = x$, so it is symmetric automatically.

Hence $\pi(\mathrm{d}x)K(x, \mathrm{d}y) = \pi(\mathrm{d}y)K(y, \mathrm{d}x)$, and $\pi K = \pi$. Note: K has no density (the atom), which is why we state detailed balance for measures.

Steps (ii)–(iii): irreducibility and aperiodicity

Proposition

If $w(x) > 0$ and $q(y \mid x) > 0$ for all x, y (e.g. Gaussian random walk $y = x + \sigma\varepsilon$), the MH chain is π-irreducible and aperiodic.

Proof. Then $\alpha(x, y) > 0$ for all x, y. For any set A with $\pi(A) > 0$ (so A has positive volume),

$$K(x, A) \;\geq\; \int_A q(y \mid x)\,\alpha(x, y)\,\mathrm{d}y \;>\; 0 \qquad \text{for every } x.$$

Irreducible: every such set is reached in one step from anywhere. *Aperiodic:* a cycle $E_1 \to E_2 \to \cdots \to E_d \to E_1$ ($d \geq 2$) would need $K(x, E_1) = 0$ for $x \in E_1$, which is impossible. □

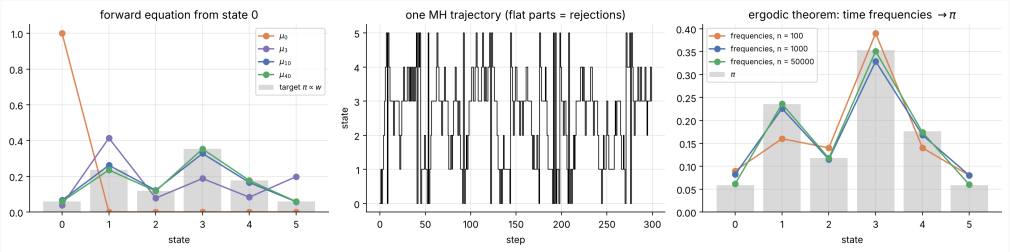
- MH with a π-irreducible kernel is also Harris recurrent (Tierney 1994): convergence from *every* start.
- Discrete case: a proposal connecting all states, plus some rejection ($p_{xx} > 0$), suffices.

Conclusion: all three hypotheses hold; MH averages converge to $\mathbb{E}_\pi f$. The *speed* depends on q (next slides).

Discrete example: an arbitrary distribution on six states

Target $w = (1, 4, 2, 6, 3, 1)$; proposal: step left or right on a ring with probability $\frac{1}{2}$.

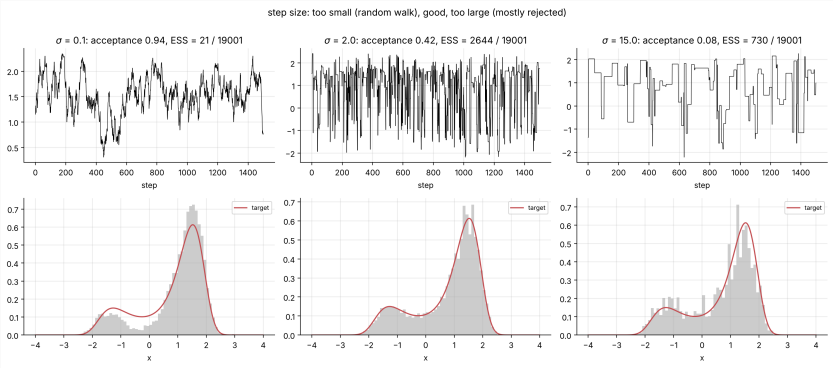
$$p_{i,i\pm1} = \tfrac{1}{2}\min\{1, w_{i\pm1}/w_i\}, \qquad p_{ii} = 1 - p_{i,i-1} - p_{i,i+1}.$$



$\pi P = \pi$ and flows symmetric (checked); $P^3 > 0$; rate = second eigenvalue 0.78. Metropolis for Markov's π with "switch letter": $P = \left(\begin{smallmatrix} 0.24 & 0.76 \\ 1 & 0 \end{smallmatrix}\right)$, $\theta = -0.76$.

Continuous example: random-walk Metropolis

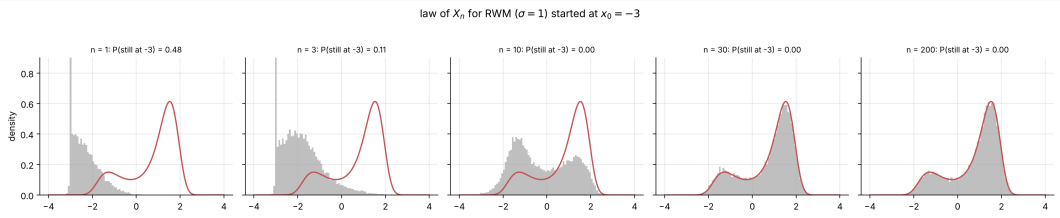
Target $\pi(x) \propto \exp\left(-x^4/4 + x^2 + 0.5x\right)$ (asymmetric, bimodal); proposal $y = x + \sigma\varepsilon$.



Step size trade-off: small $\sigma \Rightarrow$ high acceptance but a slow random walk; large $\sigma \Rightarrow$ most proposals rejected.

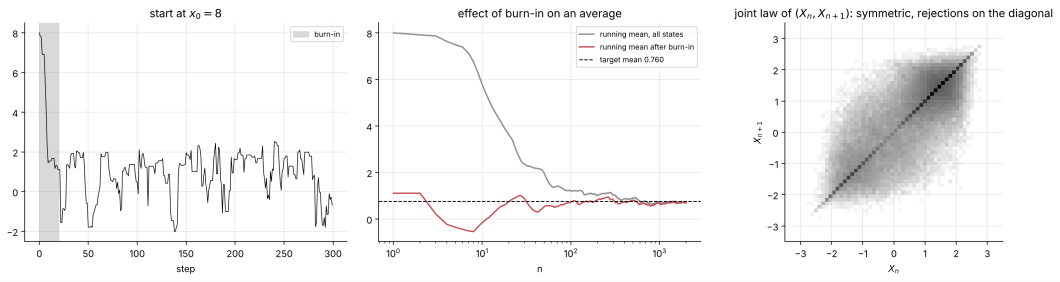
Transformation of the initial distribution under MH

20 000 chains started at $x_0 = -3$, $\sigma = 1$:



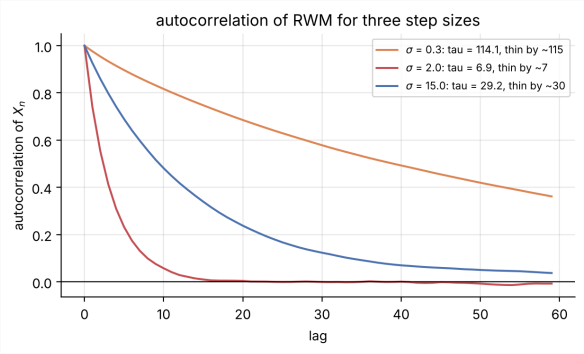
- The spike at -3 after one step is the **atom**: 48% of first proposals rejected.
- Mass first fills the nearby left mode, then crosses into the larger right mode.
- Mean of X_n: $-3 \to -0.23$ ($n = 10$) $\to 0.70$ ($n = 30$) $\to 0.76$ (target).

Burn-in, detailed balance and the atom



- Started at $x_0 = 8$: a short burn-in removes the start-up bias of the running mean.
- Joint histogram of (X_n, X_{n+1}) is **symmetric** (detailed balance). The diagonal line $X_{n+1} = X_n$ holds exactly the rejections: its mass 0.39 equals the rejection rate.

Independence: autocorrelation and ESS



- MH states are positively correlated: $\tau > 1$.
- Report $\mathrm{ESS} = n/\tau$ for each quantity.
- Tune σ to minimize τ (in high dimension: acceptance ≈ 0.234).
- Thin by $\approx \tau$ only to save storage.

Summary

1. A Markov chain is the simplest model of dependence: $p(x_0, \dots, x_n) = p_0(x_0) \prod_k p(x_{k-1}, x_k)$.
2. The law evolves by $\mu_{n+1} = \mu_n P$ (or $\int p_n(x) k(x, y)\, \mathrm{d}x$); the stationary π is its fixed point and the marginal of a typical state.
3. Detailed balance (flows balanced pair by pair) $\Rightarrow$ stationarity; it is local and needs only ratios of π.
4. Ergodic theorem: irreducible + aperiodic (e.g. $p_{ij} > 0$, $k(x, y) > 0$) $\Rightarrow$ convergence from any start and the LLN, *without independence* (Markov 1906).
5. Burn-in removes the start; dependence inflates (or deflates) the error by $\sqrt{\tau}$; ESS $= n/\tau$.
6. Metropolis–Hastings: propose, accept with $\min\{1, \frac{w(y)q(x|y)}{w(x)q(y|x)}\} \Rightarrow$ a chain for *any* $\pi \propto w$.

Notebook: `markov_chains_mh.ipynb` reproduces every figure. **Next:** Langevin, HMC, diffusions.